

The Contrast-Source Volume Integral Equation, FFT Acceleration, and the Seismic Grid T-Matrix

Background, methods, and references for the discrete-cell scattering programme

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Abstract

This note collects the methodological background and the primary literature for the discrete-cell scattering programme: the volume integral equation in its contrast-source form, the inversion framework that grew from it, the fast-Fourier-transform acceleration that makes it tractable in three dimensions, and its seismological realisation as a grid T-matrix. The contrast-source idea of van den Berg and Abubakar recasts scattering and inversion in terms of an induced source on the heterogeneity rather than the total field. The CG-FFT, adaptive integral, and precorrected-FFT methods reduce the dense integral operator to a Toeplitz convolution evaluated by FFT. In seismology, Jakobsen and collaborators built forward modelling and waveform inversion directly on the Lippmann-Schwinger T-matrix, with renormalised scattering series for strong contrast and distorted-Born iterative T-matrix inversion. The closing section places the present cubic-cell T-matrix codebase within that lineage.

Companion series. *One of a companion series on elastic multiple scattering from cubic heterogeneities on a depth-stratified background, the T-matrix programme $T = T_0(I - G_0T_0)^{-1}$: (I) the stratified inter-site propagator, the same-depth convergence theory, the SH and P/SV traction kernels, interplane side scattering and the first-shell near stencil (`EwaldIntraPlanePropagator`); (II) the contrast-source volume integral equation and the CG-FFT / seismic grid T-matrix lineage (`ContrastSourceVIE`); (III) the directional partial-wave sweep matvec and the Krylov multiple-scattering solve (`DirectionalSweepSolver`); with the empirical companion `RQ1_Planar-Decomposition-Proposal`. This is note (II).*

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1 The volume integral equation and the contrast source

Let a heterogeneity occupy a bounded domain D embedded in a known background with Green's operator G_0 . Writing the total field as incident plus scattered, the field on D obeys the Lippmann-Schwinger volume integral equation,

$$u(\mathbf{r}) = u^{\text{inc}}(\mathbf{r}) + \int_D G_0(\mathbf{r}, \mathbf{r}') \chi(\mathbf{r}') u(\mathbf{r}') dV', \quad \mathbf{r} \in D, \quad (1)$$

where χ is the *contrast* (the local deviation of the medium from the background; for the scalar wave problem $\chi \propto k^2 - k_0^2$, for elastodynamics the density-and-modulus contrast operator δL). The scattered field at receivers S is the radiation of the same induced quantity,

$$u^{\text{sc}}(\mathbf{r}) = \int_D G_0(\mathbf{r}, \mathbf{r}') \chi(\mathbf{r}') u(\mathbf{r}') dV', \quad \mathbf{r} \in S. \quad (2)$$

The contrast source. Following van den Berg and Kleinman [14], introduce the *contrast source* (the induced polarisation, or equivalent current)

$$w(\mathbf{r}) = \chi(\mathbf{r}) u(\mathbf{r}), \quad \mathbf{r} \in D. \quad (3)$$

The integral equation (1) and the data relation (2) then split into a *state (object) equation* on D and a *data equation* on S ,

$$w = \chi u^{\text{inc}} + \chi G_D w \quad \text{on } D, \quad f = G_S w \quad \text{on } S, \quad (4)$$

where G_D and G_S are the background Green's operator mapping a source on D to the field on D and on S respectively, and $f = u^{\text{sc}}|_S$ is the measured data. The whole nonlinearity of scattering now sits in the single product $\chi G_D w$.

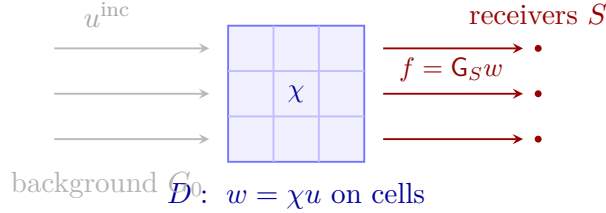


Figure 1: The volume integral equation in contrast-source form. The incident field u^{inc} induces a contrast source $w = \chi u$ on the discretised heterogeneity D ; the data f are its radiation through the background operator G_S . The state equation couples the cells of D through G_D .

2 Contrast source inversion

The contrast-source split is the basis of the *contrast source inversion* (CSI) method of van den Berg and Abubakar [14, 15, 13]. The inverse problem is to recover χ from multi-experiment data $\{f_j\}$. CSI treats the contrast χ and the contrast sources $\{w_j\}$ (one per illumination j) as *independent* unknowns and minimises a two-term normalised cost functional,

$$F(\chi, \{w_j\}) = \frac{\sum_j \|f_j - G_S w_j\|_S^2}{\sum_j \|f_j\|_S^2} + \frac{\sum_j \|\chi u_j^{\text{inc}} - w_j + \chi G_D w_j\|_D^2}{\sum_j \|\chi u_j^{\text{inc}}\|_D^2}, \quad (5)$$

the *data error* plus the *state error*. Minimisation alternates: with χ fixed, each w_j is updated by one conjugate-gradient step on (5); with the w_j fixed, χ follows in closed form because F is quadratic in χ . The decisive feature is that *no forward problem is solved at any iteration*: the expensive operator inversion of classical Born or Newton schemes is replaced by cheap gradient updates that drive the state error to zero only as the iteration converges.

Regularisation. Robustness to noise is obtained without a hand-tuned weight by the multiplicatively regularised variant, MR-CSI, in which a weighted total-variation factor multiplies (5) and adapts each iteration, preserving edges while suppressing noise [2]. CSI and MR-CSI have been carried across acoustic, electromagnetic, and elastic settings, in two and three dimensions, including finite-difference contrast-source variants for inhomogeneous backgrounds [1]; the book of van den Berg [12] is the comprehensive reference.

3 FFT acceleration of the integral operator

Every forward application of (1) or (4) requires the operator product $G_D w$, a discrete convolution of the background Green’s function against the contrast source over D . On a regular grid this operator is *block-Toeplitz*, because $G_0(\mathbf{r}, \mathbf{r}')$ depends only on $\mathbf{r} - \mathbf{r}'$, so the product is a convolution evaluated by the fast Fourier transform in $O(N \log N)$ rather than the $O(N^2)$ of a dense matrix-vector product. Paired with a Krylov solver (conjugate gradient, BiCGStab, GMRES), this is the *CG-FFT* method [11]. The three-dimensional *weak form* of Zwamborn and van den Berg [16] tests and expands the equation so that the singular kernel is integrated correctly while the Toeplitz structure, hence the FFT, is retained.

For geometries that do not fall on a single regular grid, the same FFT speed is recovered by projecting the interactions onto an auxiliary regular grid and correcting the near field directly. This is the *adaptive integral method* (AIM) of Bleszynski, Bleszynski, and Jaroszewicz [3] and the *precorrected-FFT* (pFFT) method of Phillips and White [10]. Both decompose the operator into a smooth far part, carried by FFT on the auxiliary grid, and a sparse near part, computed and corrected in real space. They are the FFT-based cousins of the fast multipole method, which instead accelerates the far field through multipole expansions.

Method	Mesh	Far field via	Near field
CG-FFT	regular grid	FFT (Toeplitz convolution)	exact on grid
AIM	arbitrary	FFT on auxiliary grid (moment match)	direct correction
pFFT	arbitrary	FFT on auxiliary grid (precorrection)	direct precorrection
FMM	arbitrary	multipole expansions	direct

Table 1: FFT-based and multipole accelerations of the dense integral operator. The first three reduce the matrix-vector product to $O(N \log N)$ by exploiting grid translation invariance; all treat the near field separately from the smooth far field.

4 The seismic grid T-matrix

In seismology the same volume integral equation is most often written through the *T-matrix*. Let V be the scattering potential (the perturbation operator of the heterogeneity relative to the background G_0). The medium T-matrix resums the multiple scattering of the whole object,

$$T = V(I - G_0 V)^{-1}, \quad G = G_0 + G_0 T G_0, \quad (6)$$

so that the Born series $T = V + V G_0 V + V G_0 V G_0 V + \dots$ is its weak-contrast expansion. Discretised on a grid of cells, V is block-diagonal (local) and G_0 is the background propagator; this is exactly the discrete-cell programme. The elastic T-matrix for a flaw or inclusion dates to Gubernatis, Domany, and Krumhansl [4], and Jakobsen and collaborators built seismic forward modelling and inversion directly on (6): the T-matrix for effective-medium and rock-physics modelling [6], acoustic forward modelling on a grid [5], and frequency-domain full-waveform inversion by direct iterative T-matrix methods [7].

Strong contrast. The Born series in (6) converges only when $\|G_0 V\| < 1$, that is for weak contrast. For strong velocity contrasts the series must be resummed. Jakobsen and Wu [8] introduced a renormalised scattering series that converges where the bare series diverges; the convergent Born series of Osnabrugge, Leedumrongwatthanakun, and Vellekoop [9] achieves the same end by preconditioning the Lippmann-Schwinger equation with tuned absorption. Either way the regular-grid Toeplitz structure of G_0 , hence the FFT acceleration of Section 3, is preserved.

Inversion. The T-matrix yields the Fréchet derivatives of the data with respect to the contrast in closed form, which is the basis of the distorted-Born iterative T-matrix (DBIT) inversion of Jakobsen and Ursin [7]: a sequence of linearised updates of V about an evolving background, each reusing the same T-matrix machinery. This is the seismological sibling of the contrast source inversion of Section 2; both avoid a full forward solve per iteration, CSI by carrying the contrast sources as unknowns, DBIT by the analytic T-matrix sensitivities.

5 Placement of the present programme

The cubic-cell T-matrix codebase is a specific, physically derived instance of this lineage. Its global solve $T = T_0 (I - G_0 T_0)^{-1}$ is the Foldy-Lax form of (6), with the bare per-cell contrast V replaced by the single-cell T-matrix T_0 that already resums the intra-cell scattering. Three choices distinguish it from the generic voxel volume integral equation:

- **An analytic cell T-matrix.** T_0 is computed in closed form, from the cube Eshelby tensor with the $r = 0$ delta-function correction up to the 27-component Galerkin representation under O_h symmetry, rather than approximated by a point or piecewise-constant contrast. The self-term singularity of the volume integral equation is thereby resolved inside T_0 .
- **A layered background.** G_0 is the stratified (Kennett) propagator, not a homogeneous whole-space Green’s function, so the depth structure of the medium is carried exactly while only the lateral contrast is discretised.
- **FFT-accelerated Foldy-Lax.** The inter-cell coupling on the regular cubic grid is block-Toeplitz and is applied by FFT inside a GMRES solve, exactly the CG-FFT acceleration of Section 3.

In the language of this note, the programme is a contrast-source volume integral equation with an analytic, physically exact cell T-matrix, an FFT-accelerated forward operator, and a layered background, placed squarely in the seismic grid T-matrix tradition of Gubernatis and of Jakobsen while contributing the closed-form cubic-cell scatterer.

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